

Variance reduction

Covered by chapter 6.4-6.6 in the textbook.

For the various Monte Carlo integration techniques we have considered so far we have seen that the standard deviation of the estimator is on the form $K/\sqrt{n}$ where K is some constant.

Thus:

- Precision increases by $\sqrt{n}$.
- Increasing the precision has a high computational cost.

Are there other ways of reducing the standard deviations/variance of the Monte Carlo estimators?

We can not get around the $\sqrt{n}$, but we shall see that there are several ways to decrease K .

Antithetic variables

First recall from chapter 2 that if $Y = \sum_{i=1}^n a_i X_i + b$, then

$$\text{Var}(Y) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=1}^{i-1} a_i a_j \text{Cov}(X_i, X_j).$$

For the average of two random variables X_1 and X_2 , $Y = \bar{X} = \frac{1}{2}X_1 + \frac{1}{2}X_2$, this implies:

$$\begin{aligned} \text{Var}\left(\frac{X_1 + X_2}{2}\right) &= \frac{1}{4} \text{Var}(X_1) + \frac{1}{4} \text{Var}(X_2) \\ &\quad + \frac{2}{4} \text{Cov}(X_1, X_2) \\ &< \frac{\text{Var}(X)}{2} \quad \text{if } \text{Cov}(X_1, X_2) < 0 \end{aligned}$$

I.e. the variance of the average is lower than the variance of the average of two independent variables if the covariance (and thus the correlation) is negative.

This observation can be used to reduce the variance in MC integration estimators.

How to simulate negatively correlated variables?

If $U \sim \text{U}[0, 1]$ then $V = 1 - U$ is also $\text{U}[0, 1]$, and U and V will be negatively correlated.

This extends to:

- If $X = F^{-1}(U)$ and $Y = F^{-1}(1 - U)$ then X and Y are r.v.s from the same distribution with cdf F , and they are negatively correlated.
- If $U \sim \text{U}[a, b]$ then $V = a + b - U$ is also $\text{U}[a, b]$, and U and V will be negatively correlated.

Variables created as above are called *antithetic*.

MC integration with antithetic variables

Consider now estimating

$$\theta = \int_a^b g(x)dx$$

by

$$\hat{\theta}_{CMC} = \frac{b-a}{n} \sum_{i=1}^n g(U_i)$$

Intuitively, if the negative correlation between a pair of antithetic random variables U_i and V_i extends to negative correlation between $g(U_i)$ and $g(V_i)$ we could reduce the variance in the above estimator if we replace the iid uniform random variables with antithetic variables.

It can be shown (details are given in the book) that **if $g(x)$ is monotonic on the interval $[a, b]$ then $\text{Cov}(g(U_i), g(V_i)) < 0$ when $\text{Cov}(U_i, V_i) < 0$.**

Thus if $g(x)$ is **monotonic** on $[a, b]$ we can estimate the integral by

$$\hat{\theta}_{AT} = \frac{b-a}{n} \left(\sum_{i=1}^{n/2} g(U_i) + \sum_{i=1}^{n/2} g(V_i) \right)$$

where $U_1, U_2, \dots, U_{n/2}$ are iid $U[a, b]$ and $V_i = a + b - U_i$ for $i = 1, 2, \dots, n/2$.

Example:

Compare crude and antithetic MC for estimating:

$$\int_a^b \cos(x) dx$$

See R code: `Variance_reduction.R`

General basic Monte Carlo integration with antithetic variables

For

$$\theta = \int_A g(x)dx$$

where $g(x)$ can be decomposed into $g(x) = h(x)f(x)$ where $f(x)$ is a probability density on the set A such that

$$\theta = \int_A h(x)f(x)dx = E(h(X))$$

Then **if $h(x)$ is monotonic on A** a Monte Carlo estimate using antithetic variables is

$$\hat{\theta}_{GAT} = \frac{1}{n} \sum_{i=1}^{n/2} (h(X_i) + h(Y_i))$$

where $X_i = F^{-1}(U_i)$ and $Y_i = F^{-1}(1 - U_i)$ for $i = 1, 2, \dots, n/2$ and $U_1, U_2, \dots, U_{n/2}$ are iid $U[0, 1]$.

If your f is $N(\mu, \sigma^2)$, then an antithetic pair is obtained as $X_i = \mu + \sigma Z_i$, $Y_i = \mu - \sigma Z_i$ where $Z_i \sim N(0, 1)$.

Example:

Find $\int_0^\infty \sqrt{x} e^{-4x} dx$

Recall similar trick earlier;

$$h(x) = \frac{\sqrt{x}}{4}, \quad f(x) = 4 \exp(-4x) \Rightarrow h(x)f(x) = g(x)$$

The proposal f is exponentially distributed,
hence sampling f using inverse transform
sampling amounts to the antithetic pair: $X_i =$
 $-\log(1 - U_i)/4$ and $Y_i = -\log(U_i)/4$

See R code: `Variance_reduction.R`

Importance sampling

Recall the general MC approach to approximate

$$\theta = \int_A g(x) dx$$

If $g(x)$ can be decomposed into $g(x) = h(x)f(x)$ where $f(x)$ is a probability density on the set A , then

$$\theta = \int_A h(x)f(x)dx = E(h(X))$$

where X is a random variable with density $f(x)$. Then a general crude Monte Carlo estimate is

$$\hat{\theta}_{GCMC} = \frac{1}{n} \sum_{i=1}^n h(X_i)$$

where $X_1, X_2 \dots X_n$ are iid from f .

The variance of this estimator is

$$\text{Var}(\hat{\theta}_{GCMC}) = \frac{1}{n} \text{Var}(h(X))$$

We can alternatively write this as

$$\theta = \int_A g(x)dx = \int_A \frac{g(x)}{f(x)} f(x)dx = \mathbb{E} \left(\frac{g(X)}{f(X)} \right)$$

where $f(x)$ is a density function on A .

Importance sampling is about choosing the density $f(x)$ smart such that the variance of the estimate gets as small as possible.

$$\hat{\theta}_{IS} = \frac{1}{n} \sum_{i=1}^n \frac{g(X_i)}{f(X_i)}$$

where $X_1, X_2 \dots X_n$ are iid from f .

The variance of this estimator is

$$\text{Var}(\hat{\theta}_{GCMC}) = \frac{1}{n} \text{Var} \left(\frac{g(X)}{f(X)} \right)$$

The variance is smaller the flatter $g(x)/f(x)$ is. I.e. we should pick a function $f(x)$ which is close to $g(x)$ in shape (where $f(x)$ has significant mass) and which is easy to sample from.

Example: Find $\int_{-1.5}^{1.5} \cos(x) dx$

See R code: `Variance_reduction.R`

Notice that the support of the density $f(x)$ used in importance sampling must cover at least all of A , but we can also use densities with larger support. If the support is larger than A we just rewrite the integral with indicator functions:

$$\begin{aligned}\theta &= \int_A g(x)dx = \int_B \frac{g(x)I(x \in A)}{f(x)} f(x)dx \\ &= \mathbb{E} \left(\frac{g(X)I(X \in A)}{f(X)} \right)\end{aligned}$$

where $A \subset B$ and B is the support of f .

For efficient sampling the probability of drawing values outside A should be small. I.e. the density function $f(x)$ should be “small” outside A .

Example: Find $\int_0^3 \frac{e^{-x}}{1+x^2} dx$

Using three types of proposals:

- standard exponential:
 $f(x) = \exp(-x)$, $x > 0$ and indicator trick
- truncated exponential:
 $f(x) = \exp(-x)/(1 - \exp(-3))$, $0 < x < 3$.
Simulated as
 $X_i = -\log(1 - U_i(1 - \exp(-3)))$ where
 $U_i \sim \text{uniform}(0,1)$.
- truncated exponential with antitetics.

See R code: `Variance_reduction.R`

Tail behavior

A premiss for successful importance sampling is that the importance sampling weight $g(x)/f(x)$ remains bounded as $|x| \rightarrow \infty$. Unboundedness typically stem from that $f(x)$ goes too fast to zero; "thin tails".

In many applications of importance sampling, a normal distribution is used as the importance density. This **may** be problematic as the normal distribution has thin tails.

Diagnostic: Check the distribution of your importance weights $g(X_i)/f(X_i)$, $i = 1, \dots, n$. If some weights are "much" larger than the rest, there may be trouble...

See R code: `Importance sampling.R`

Second order approximation

In high-dimensional cases, i.e. to calculate

$$\theta = \int g(\mathbf{x}) d\mathbf{x} \text{ where } \mathbf{x} \in \mathbb{R}^d,$$

for $g(\mathbf{x})$ not too far from a multivariate normal distribution, one common approach for importance sampling is to use the multivariate normal proposal $f(\mathbf{x})$ with mean

$$\mu = \arg \max_{\mathbf{x}} \log g(x)$$

and covariance matrix being the inverse negative Hessian of $\log g$ evaluated at μ , i.e.

$$\mathbf{G}_{i,j}(\mathbf{x}) = -\frac{\partial^2}{\partial x_i \partial x_j} \log g(\mathbf{x}), \quad i, j = 1, \dots, d$$

and

$$\Sigma = (\mathbf{G}(\mu))^{-1}$$

Both μ and Σ may be obtained using the `optim`-function in R:

See R code: `Importance sampling.R`

State of the art importance sampling

Current state of the art importance sampling implementations relies on f being a mixture of multivariate student- t distributions.

Such distributions are very flexible and may therefore approximate well more or less any shape of g . However, determining the parameters of the mixture is a complicated optimization problem.

If you are interested; have a look at R-package `AdMit`

Read on your own: Control variables

We still consider integrals which can be interpreted as expectations, i.e. $\theta = E(g(X))$.

Suppose now that there exist a function h such that $\mu = E(h(X))$ is known and $g(X)$ is correlated with $h(X)$. Then for

$$\hat{\theta}_c = g(X) + c(h(X) - \mu)$$

we have that for any c

$$\begin{aligned} E(\hat{\theta}_c) &= E(g(X)) + c(E(h(X)) - \mu) = \theta \\ \text{Var}(\hat{\theta}_c) &= \text{Var}(g(X)) + c^2 \text{Var}(h(X)) \\ &\quad + 2c \text{Cov}(g(X), h(X)) \end{aligned}$$

As a function of c , $\text{Var}(\hat{\theta}_c)$ is minimized when

$$c = c^* = \frac{-\text{Cov}(g(X), h(X))}{\text{Var}(h(X))}$$

and the minimum variance is

$$\text{Var}(\hat{\theta}_c^*) = \text{Var}(g(X)) - \frac{\text{Cov}(g(X), h(X))^2}{\text{Var}(h(X))}$$

The variable $h(X)$ is called a *control variable*.

The reduction in variance is

$$\frac{\text{Cov}(g(X), h(X))^2}{\text{Var}(h(X))}$$

and the relative reduction is

$$\frac{\text{Cov}(g(X), h(X))^2}{\text{Var}(h(X))\text{Var}(g(X))} = \text{Cor}(g(X), h(X))^2$$

I.e. the stronger the correlation between $g(X)$ and $h(X)$ is the better.

To calculate c^* we need $\text{Cov}(g(X), h(X))$ and $\text{Var}(h(X))$. If these are unknown they are estimated by a preliminary simulation.

The final estimate of the integral is

$$\hat{\theta}_{CV} = \overline{g(X)} + c^*(\overline{h(X)} - \mu)$$

Antithetic variable estimators can be considered as a special cases of control variables - see the book.

Example:

Find

$$\int_0^{1.5} \cos(x) dx$$

See R code: `Variance_reduction.R`